

Chapter 1.

Bivariate data.

Situation.

A statistician was asked to investigate whether countries with higher levels of consumption of a particular foodstuff tended to have higher death rates due to heart disease. Searching data bases available from various groups concerned with international health the statistician compiled the following table:

	CONSUMPTION OF THE FOODSTUFF Kilograms per person per year.	DEATHS DUE TO HEART DISEASE Per year per 100 000 of population.
Country A	57	299
Country B	20	124
Country C	13	154
Country D	38	156
Country E	51	152
Country F	48	295
Country G	58	264
Country H	35	208
Country I	29	250
Country J	8	84
Country K	28	158
Country L	21	175
Country M	10	120
Country N	40	220
Country O	41	251
Country P	46	202
Country Q	53	248
Country R	64	290
Country S	66	296
Country T	47	268

What would be your conclusion? (Justify your decision.)

	Party A	Party B	Totals
Male	0.51	0.25	0.76
Female	0.10	0.14	0.24
Totals	0.61	0.39	1.00

	Party A	Party B	Totals
Male	51%	25%	76%
Female	10%	14%	24%
Totals	61%	39%	100%

Now observations like

- ☞ 51% of all of the MPs from the two parties are males from Party A,
 - ☞ 76% of all the MPs from these two parties are male,
- more readily come to mind.

Row and column percentages.

We could also consider percentages of the row totals, called row percentages, as shown below left, and percentages of column totals, called column percentages, as shown below right. (Check that you agree with the various entries in each table.)

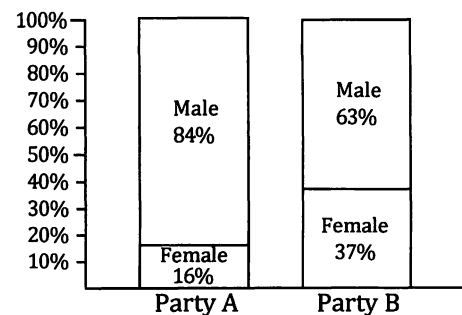
	Party A	Party B	Totals
Male	67%	33%	100%
Female	41%	59%	100%

	Party A	Party B
Male	84%	63%
Female	16%	37%
Totals	100%	100%

Now observations like

- ☞ 67% of the male MPs from these two parties are from Party A,
 - ☞ 59% of the female MPs from these two parties are from Party B,
 - ☞ 37% of Party B MPs are female compared with 16% of Party A MPs,
- are more apparent.

Displaying the percentages from the table above right, as two proportionally divided columns, gives the graph shown on the right (sometimes referred to as a *stacked 100% column graph*). We can see from this diagram, as indeed we could from the table, that as we move from the Party A column to the Party B column the male-female proportions change noticeably. The proportion of MPs who are female is much greater in Party B than in Party A. I.e., with regard to gender balance it does matter which party we are considering. This suggests there is an *association* between the two categorical variables of Party and Gender of MP. If there were no association we would not expect the proportions to change so noticeably as we move from the Party A column of our graph to the Party B column.



Hint: For ease of dividing each column into percentages, when drawing the column graph by hand make the column height 100 mm.

Choosing the Australian Federal Government, or one of the State or Territory Governments, research the number of male and female MPs in the major parties and present two way tables, proportional tables and percentage tables, similar to those shown here, for your data.

Make some comments based on your findings.

Suppose instead that we investigated the mode of transport to work (bus, car, train, other) and gender. Further suppose that survey results gave rise to the following table:

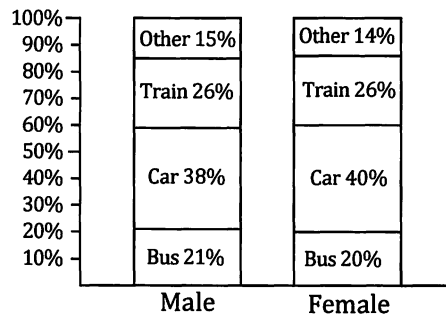
	Bus	Car	Train	Other	Totals
Male	26	47	32	19	124
Female	35	70	45	24	174

This table gives rise to the table below showing row percentages, given to the nearest whole percentage. (Check that you agree with the percentages).

	Bus	Car	Train	Other	Totals
Male	21%	38%	26%	15%	100%
Female	20%	40%	26%	14%	100%

Notice that the male-female proportions are very similar in each column, suggesting the lack of any association between gender and mode of transport to work. I.e., with regard to mode of transport to work it does not matter what gender the person is.

The proportional column graph shown below similarly indicates that it does not matter whether we are considering male or female, the proportions using the various modes of transport are almost identical as we move from one gender column of the graph to the other.



The proportions of the various modes of transport are not influenced by what gender we are considering. Contrast this with the gender balance of MPs considered earlier. The proportion of female MPs was certainly influenced by which party we were considering.

If in changing the category of one variable (eg changing from Party A to Party B or from male to female) the proportions in the various categories of the other variable change, then this suggests that the two variables are **associated**. Of course we would expect some small changes to occur just because of the variation in the samples involved so for the existence of an association to be strongly suggested we would need the changes in proportions to be significant.

In the MPs situation we looked at whether considering a different party meant that the proportion of male MPs to female MPs changed. To do this we split the party numbers up into male and female percentages.

In the modes of travel situation we looked at whether considering a different gender meant that the proportion of modes of transport changed. To do this we split the male and female numbers up into types of travel percentages.

As well as being able to construct and examine appropriate two way tables and graphs to see if association is suggested we should also be able to describe an association by commenting on the nature and magnitude of any changes in proportions that exist. The next two examples, and the exercise that follows, cover these aspects.

Example 1

A number of men were asked which was their most preferred type of movie out of:
Crime, Romance or Horror.

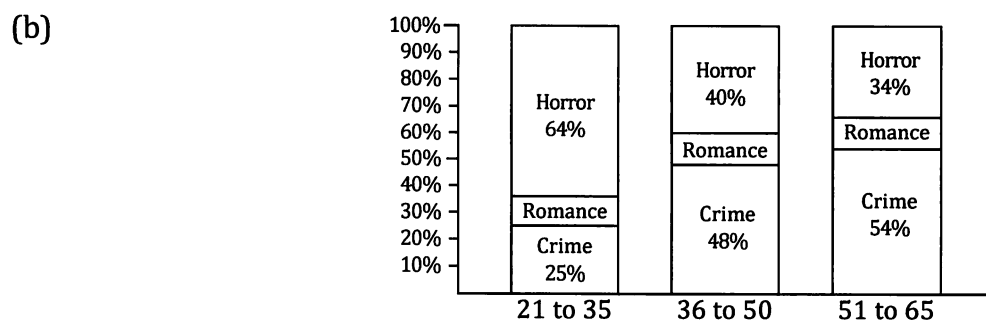
The data obtained was sorted according to the age of the respondent and gave rise to the table shown below.

		Film Type		
		Crime	Romance	Horror
Age in Years	21 to 35	26	11	67
	36 to 50	35	9	29
	51 to 65	22	5	14

- Draw the frequency table showing row percentages (i.e. % based on row totals).
- Draw a proportional column graph involving three columns of equal height with one column for each of the three age groups.
- Comment on whether any association between the variables is suggested and write some comments describing the association.

(a)

		Film Type			Totals
		Crime	Romance	Horror	
Age in Years	21 to 35	25%	11%	64%	100%
	36 to 50	48%	12%	40%	100%
	51 to 65	54%	12%	34%	100%



- (c) The ages of 36 to 50 and 51 to 65 show similar proportions of preferred film type but there is a clear difference in the proportions in the first column to those in the other two, suggesting an association between *age* and *preferred film type*.
The percentage of each age category preferring romance films is virtually unchanged at about 12%, irrespective of what age category we are considering. Whilst about two thirds of 21 to 35 year old males prefer horror movies only one third of 51 to 65 year old males have horror movies as their preferred type. Having crime as the preferred choice goes the other way with just one quarter of 21 to 35 year old males having it as their preferred type whilst just over half of 51 to 65 year olds choose it as their preferred type of movie.

In part (c) of the previous example the comments given, after the first, are given by way of example. There are other correct comments that could be given.

Example 2

Classifying people as one of the three categories:

Left handed, Right handed, Mixed / Ambidextrous,
a survey considering whether the "handedness" of a firstborn child could be explained by the "handedness" of the mother, produced the following frequency table.

		Handedness of Mother		
		Left handed	Right handed	Mixed / Ambi
Handedness of firstborn child	Left handed	4	21	1
	Right handed	21	155	3
	Mixed / Ambi	2	8	1

Redraw the above table showing column percentages and use it to identify and comment on any patterns shown and whether any association between the handedness of a child and the handedness of the mother is suggested.

The table with column percentages (nearest percent) will be as shown below.

		Handedness of Mother		
		Left handed	Right handed	Mixed / Ambi
Handedness of firstborn child	Left handed	15%	11%	20%
	Right handed	78%	84%	60%
	Mixed / Ambi	7%	4%	20%
Totals		100%	99%*	100%

* The total of 99%, rather than 100%, is simply due to rounding.

Moving across the columns in the above table does show some variation of proportions, the most noticeable difference being in that of the proportions in the "Mixed/Ambi" column compared to the others. However the percentages for that column are based on a very small total (5) and a change of just 1 of these 5 from one category to another would greatly change the percentages. Hence this final column is of limited use.

Considering the changes in the proportions in the first two columns the changes are not huge and could simply be due to the sample of people involved. Certainly a greater percentage of the right handed mothers had a right handed firstborn child (84%) than did left handed mothers (78%) but the difference in these percentages is only 6%. Similarly a higher percentage of left handed mothers had left handed firstborns (15%) than did right handed mothers (11%).

Thus the table suggests that whilst the handedness of the mother and the handedness of the firstborn could, perhaps, to some limited extent, be associated with each other the association is not particularly strong (and there were only 4 left handed firstborns of left handed mothers involved in the survey).

Interestingly, the table above makes no mention of the handedness of the child's father. If we wanted to investigate whether there is some association between a child's handedness and that of the parents this aspect could be an important consideration.

Note: In this example the proportional column graph, though not requested, could also be drawn to assist interpretation if so desired.

Note: Tables given in this chapter have been created for the purposes of this book. They are not necessarily taken from real data but do attempt to reflect a situation that could exist.

Exercise 1A

1. Suppose that a survey asking a number of people their age and which they most preferred as a main course at a restaurant, out of beef, chicken, fish or other, gave rise to the two way table shown below.

	Beef	Chicken	Fish	Other
18 - 30	10	7	3	1
31 - 43	9	10	12	4
44 - 56	7	17	15	3
Over 56	2	5	8	2

- How many people responded to the survey?
 - How many of the respondents chose beef as their most preferred main course at a restaurant?
 - What percentage of those choosing beef as their most preferred course at a restaurant were aged 31 to 43?
 - What percentage of those aged 31 to 43 chose beef as their preferred main course at a restaurant?
2. Suppose that a survey investigating the gender and current marital situation of a number of adults gave rise to the two way frequency table shown below.

For current marital situation respondees indicated one of the four categories:

single (and never married), married,
divorced (and not currently married), widowed (widow or widower).

	Single	Married	Divorced	Widowed
Male	49	68	24	21
Female	60	92	30	32

- How many people does the frequency table involve?
- How many males does the frequency table involve?
- What percentage of the males involved in the survey were divorced.
- What percentage of the divorced people involved in the survey were males?
- According to the survey results which is the greater, the percentage of males who are single or the percentage of females who are single?

3.

	A	B	C	D
E	98	116	154	336
F	52	66	64	18

The two way frequency table shown above involves two variables, one having categories A, B, C and D and the other having categories E and F.

Create 3 tables, together with appropriate totals (and percentages to nearest 1%):

- A table showing each entry as a percentage of the total (904).
- A table showing row percentages. (Percentages based on row totals.)
- A table showing column percentages. (Percentages based on column totals.)

4. Suppose that a survey of the preferred football code, out of Australian Rules, Rugby League, Rugby Union and Soccer and the state where the respondent lives, was conducted for a sample of 291 people living in four states. The results obtained gave rise to the following table:

	NSW	Queensland	Victoria	WA
Australian Rules	12	15	54	39
Rugby League	50	32	4	2
Rugby Union	20	13	6	6
Soccer	15	8	8	7

- (a) Redraw the above table showing column percentages (to the nearest 1%).
- (b) Draw a proportional column graph involving four columns of equal height with one column for each of NSW, Queensland, Victoria and WA.
- (c) Make some comments summarizing some of the information shown in the percentages table and column graph and in particular comment on whether any association between the preferred football code and the state in which a person lives is suggested.
5. In addition to the usual drug treatment for a particular disorder a medical facility also invites sufferers to follow one of three other programs, A, B or C. The success or failure of 119 people treated for the disorder gave rise to the following table.

	Success	Failure
Drug program only	18	10
Drug program + program A	13	5
Drug program + program B	24	8
Drug program + program C	29	12

- (a) Redraw the above table showing row percentages (to the nearest 1%).
- (b) Draw a proportional column graph involving four columns of equal height with one column for each of the four "program categories".
- (c) Make some comments summarizing some of the information shown in the percentages table and column graph and in particular comment on whether any association is suggested between success and failure and the programs.
6. The table below shows the numbers of males and females employed by a particular organisation, grouped according to age.

	30 and under	31 to 40	41 to 50	51 to 60	61 & over
Male	47	39	62	76	15
Female	51	34	58	35	6

- (a) Redraw the above table showing column percentages (to the nearest 1%).
- (b) Investigate how the proportions of males and females change as we move across the age categories. Is any association suggested? Comment on your findings and write some summary statements describing any differences you observe in gender percentages across the various age categories.
(The organization had, in recent years, introduced a gender equality policy.)

7. A number of newly wedded couples, who did not yet have children, were asked:
If you were to have a daughter, which activity out of
Music, Dance, A Ball Sport, Gymnastics, Horse Riding
would you most like them to excel in?

The responses from 177 males and 194 females gave rise to the following table:

	Music	Dance	A Ball Sport	Gymnastics	Horse Riding
Male	62	18	53	32	12
Female	73	39	27	39	16

- (a) Redraw the above table showing row percentages (to the nearest 1%).
(b) Investigate how the proportions in each category of activity changes as we move across the gender categories. Comment on your findings and write some summary statements describing any differences you observe in activity percentages as we move across the gender categories.
8. A survey of 900 people aged 15 and over, and in full time employment, classified the employment of each person as being one of three types:

An employee.

An independent, working under contract.

A business owner selling goods to the public.

The survey gave rise to the following table:

	15 to 29	30 to 44	45 to 59	60 and over
Employee	203	183	231	99
Independent	7	17	38	24
Business owner	4	15	40	39

- (a) Redraw the above table showing column percentages (to the nearest 1%).
(b) Investigate how the proportions in each employment category change as we move across the age categories. Comment on your findings and write some summary statements describing any differences you observe in the percentages in each employment type as we move across the age categories.
9. Suppose that a number of adults were asked to rate themselves on a happiness scale:

Usually very happy

Usually happy

Neutral

Usually sad

and that the results of the survey were as given in the table below.

	Usually very happy	Usually happy	Neutral	Usually sad
Male	23	34	20	9
Female	36	52	26	11

Redraw the above table showing row percentages (to the nearest 1%) and use it to identify and comment on any patterns shown and whether any association between happiness and gender is suggested.

Explanatory variable and response variable.

The situation at the beginning of this chapter considered whether countries with higher levels of consumption of a particular foodstuff tended to have higher death rates due to heart disease. Researchers would be interested to investigate whether the higher death rates were perhaps in some way *explained* by the higher consumption of the foodstuff. In this situation we say that the consumption of the foodstuff is the **explanatory variable**, also called the independent variable or predictor variable.

Were the higher death rates due to heart disease in some way a *response* to the higher levels of consumption of the foodstuff? We say that the death rate is the **response variable**, also called the dependent variable or predicted variable.

When investigating relationships in bivariate data, the **explanatory variable** is the variable used to explain or predict a difference in the **response variable**.

Earlier in this chapter we considered whether the balance of male/female MPs could depend upon the party the MP was in.

Explanatory variable: The party the MP was in.

Response variable: Whether the MP was male or female.

We looked at whether the choice of mode of transport to work could in some way be explained by gender.

Explanatory variable: Gender.

Response variable: Choice of mode of transport to work.

We also considered whether the preferred type of movie could be explained by age.

Explanatory variable: Age

Response variable: Preferred type of movie.

We might consider whether high blood pressure increases the risk of a heart attack.

Explanatory variable: High blood pressure.

Response variable: Risk of heart attack.

If you find it difficult to identify which variable is the explanatory variable and which the response variable it can be helpful to ask:

Which variable explains differences in the other?

Which variable would be used to predict the other?

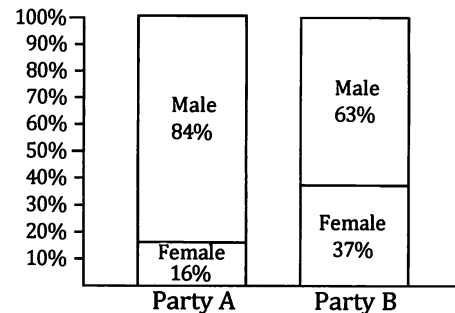
Which occurs first? As in high blood pressure (explanatory) & heart attack (response).

If we know what it is we are trying to investigate the identification of explanatory variable and response variable usually makes sense. However, if all we are given is two categorical variables it may not be obvious which of the two variables is the explanatory variable and which the response variable. Changes in one variable A may not give any explanation of the changes in variable B and variable B may not respond in any way to changes in variable A. Situations could occur where in most cases it would be logical to use variable A to predict variable B but just occasionally a situation could need us to predict variable B from variable A. However, in this chapter, if a question requires us to decide which variable is which such "unusual" situations are avoided and the identification of which variable is the explanatory variable and which is the response variable will follow what should seem to be the logical choice.

Which to use – row percentages or column percentages?

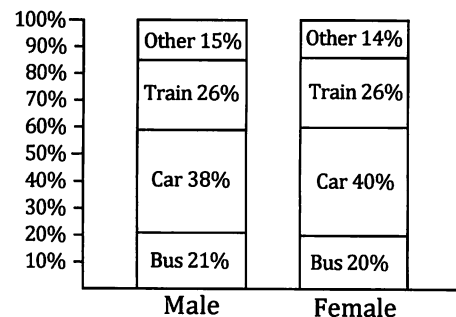
In the MPs situation encountered earlier in this chapter we considered whether the party an MP was in would explain the male-female balance. The political parties A and B formed the explanatory variable and gender (male-female) the response variable. The table gave the political parties as columns and we viewed the table of column percentages (i.e. percentages based on column totals) and created our proportional column graph from these percentages.

	Party A	Party B
Male	84%	63%
Female	16%	37%
Totals	100%	100%



In the modes of transport to work situation we considered whether gender explained any variations in choice of mode of transport to work. Now gender is the explanatory variable and mode of transport the response variable (although we then went on to show no association indicating no response to gender). The table gave gender as the rows in the table and we viewed the table of row percentages, i.e. percentages based on row totals, and created our proportional column graph from these percentages.

	Bus	Car	Train	Other	Totals
Male	21%	38%	26%	15%	100%
Female	20%	40%	26%	14%	100%



In each case the decision about which percentage table to use depended on which variable involves the categories we want to consider "moving across".

We wanted to consider male-female proportions as we moved across political parties. Thus we wanted percentages of male-female within each political party.

We wanted to consider mode of transport proportions as we moved across gender categories. Thus we wanted percentages of mode of transport within each gender.

For each situation, with thought, we could determine whether the row percentages or the column percentages were the more appropriate. However, if you prefer a rule:

- ☞ If the explanatory variable uses the rows to show its different categories use row percentages (percentages based on row totals).
- ☞ If the explanatory variable uses the columns to show its different categories use column percentages (percentages based on column totals).

I.e., in each case, split the category totals of the explanatory variable into percentages.

Example 3

A number of people were asked whether they were in favour of increasing the number of driving hours a learner driver must do before gaining a license, not in favour or were undecided on the issue. The results, by age of respondent, led to the following table.

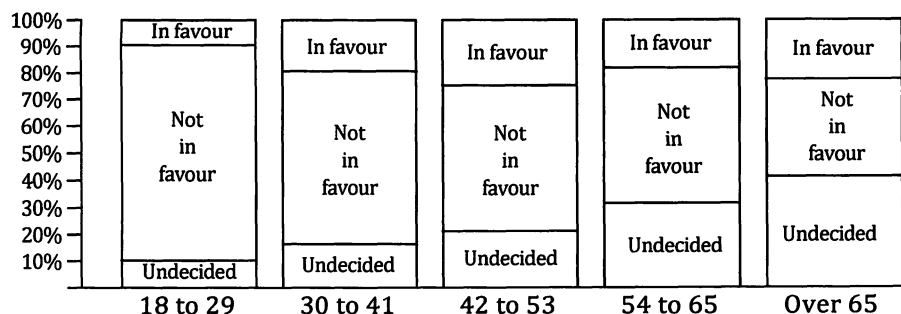
	Age of respondent (in years)				
	18 to 29	30 to 41	42 to 53	54 to 65	Over 65
In favour of more hours	3	10	12	5	5
Not in favour of more hours	24	32	26	13	8
Undecided	3	8	10	8	9

These results are to be used to investigate whether the way a person feels about this issue could be explained by the age of the person.

- Which is the explanatory variable and which the response variable?
- Draw the table showing either row or column percentages as appropriate and use your table to create a proportional column graph.
- Comment on whether there seems to be an association between the variables, explaining your reasoning and describing the association.

- We wish to consider whether the response can be explained by age. Age is the explanatory variable and the response to the question the response variable.
- We wish to see if the response percentages change as we move across the age categories. Thus we need column percentages. Or, using the rule, the explanatory variable uses columns to display its categories so we want column percentages.

	Age of respondent (in years)				
	18 to 29	30 to 41	42 to 53	54 to 65	Over 65
In favour	10%	20%	25%	19%	23%
Not in favour	80%	64%	54%	50%	36%
Undecided	10%	16%	21%	31%	41%
Totals	100%	100%	100%	100%	100%



- The change in the proportions as we move across the age categories suggests that there is an association between the response and age. Whilst in each age category the "Not in favour" is much larger than the "in favour" the size of the "not in favour" decreases as the age category increases. The percentage undecided shows fairly steady increase as age category increases, from 10% of 18 to 29s undecided up to 41% of over 65s undecided. The proportion "in favour" is least in the 18 to 29 age category (10%) which is roughly half the proportion in each of the other age categories (from 19% to 25%).

Exercise 1B

For numbers 1 to 16 identify the explanatory variable and the response variable.

1. A survey investigating whether preferred film type can be explained by the gender of the person asked to give their preference.
2. A survey investigating whether the activity a parent would most like a daughter to excel at can be explained by the gender of the parent.
3. A survey investigated whether there was any evidence that the likelihood of a person developing respiratory problems in the teenage years could be associated with the age of the person's mother at the time the person was born.
4. A survey investigated whether the left or right handedness of a child was associated with the left or right handedness of the child's mother.
5. A survey investigating whether a person's preferred football code, out of AFL, Rugby League, Rugby Union and Soccer could be explained by the state a person lives in.
6. Investigating how the grade a student gets in an exam may relate to the amount of study a student does for that exam.
7. Investigating how the number of babies born in a country per year might relate to the population of the country.
8. Investigating into how well a person sleeps at night and a rating of the person's grumpiness the next morning.
9. Whether or not a person has a particular mental disorder and whether they claim to "hear voices".
10. An investigation into how frequently a person brushes their teeth:
after most meals, once a day, 4 to 6 times a week,
 1 to 3 times a week, less than once a week.
and gender.
11. The classification of a smoker as being high (i.e. smokes a lot), medium or low and the persons respiratory function, also ranked as high, medium or low.
12. An investigation into the age of women when they have their first child and the age of each women's mother when the mother had her first child.
13. Heating milk in a saucepan with the two variables being
 the temperature of the milk
and the time the heat has been applied for.
14. Lack of sleep and level of tiredness.
15. Stress levels and heart disease.
16. How dangerous a driver is to other road users and the amount of alcohol the driver has consumed recently.

17. Suppose a survey was carried out to investigate whether there was any evidence to suggest that the likelihood of someone developing respiratory problems in their teenage years was associated with the amount that their mother engaged in a particular activity during pregnancy. Further suppose that the survey produced the following results.

		Level of respiratory problems in teenage years		
		None	Moderate	Severe
Engaged in activity during pregnancy	Not at all	63	17	5
	Sometimes	18	16	7
	Often	10	18	6

- Which is the explanatory variable and which the response variable in this situation?
 - Are the explanatory variable's categories displayed using different rows or different columns?
 - Recreate the table showing either row or column percentages as appropriate.
 - Use your table to create a proportional column graph with equal height columns each representing 100% of an appropriate category.
 - Comment on whether there seems to be an association between the variables, explaining your reasons and describing the association.
18. Could an individual's preferred soft drink be explained in any way by the age of the individual? Let us suppose that to investigate this question a number of people, aged from 16 to 65, were asked to nominate their most preferred drink out of:

A fizzy drink
Tea

Fruit Juice
Milk / Flavoured Milk

Coffee
Water

The responses led to the creation of the following table of most preferred soft drink, out of those listed above, for the 416 people who responded.

Age ↓	Number of people nominating the drink as most preferred soft drink.					
	Fizzy Drink	Fruit Juice	Coffee	Tea	Milk/ Flavoured Milk	Water
16 to 25	28	20	7	3	13	9
26 to 35	28	20	22	12	14	21
36 to 45	11	10	14	8	5	17
46 to 55	8	14	23	15	6	27
56 to 65	5	10	15	11	4	16

- Which is the explanatory variable and which the response variable in this situation?
- Recreate the table showing either row or column percentages as appropriate.
- Comment on whether there seems to be an association between the variables, explaining your reasons and describing the association.

19. In an attempt to determine what factors might explain the level of salary for a person working in a particular field, a nationwide survey of a number of people aged between 30 and 40 and in full time employment in the particular field were asked, amongst other things, their salary, their blood group and the highest level of education they had achieved. With the salaries then grouped into the categories Medium, High and Very High, the results led to the following frequency tables.

		SALARY			Totals
		Medium	High	Very High	
BLOOD GROUP	A	85	53	12	150
	B	21	17	3	41
	AB	7	6	2	15
	O	106	67	17	190
Totals		219	143	34	396

		HIGHEST LEVEL OF EDUCATION				Totals
		School	Training Diploma or Certificate	Degree	Higher Degree	
Salary	Medium	40	52	111	16	219
	High	10	25	71	37	143
	Very High	2	4	16	12	34
	Totals	52	81	198	65	396

- (a) Why do you think the survey was limited to people in full time employment and to people of a particular age group?
- (b) Use the first table to investigate whether blood group might explain the salary level of a person aged between 30 and 40 and working full time in this particular field. State which variable is the explanatory variable, which the response variable, present an appropriately percentaged version of the table, include a graph if you wish and comment on your findings.
- (c) Use the second table to investigate whether level of education might explain the salary level of a person aged between 30 and 40 and working full time in this particular field. State which variable is the explanatory variable, which the response variable, present an appropriately percentaged version of the table, include a graph if you wish and comment on your findings.

20. An organisation developed a program to help people give up smoking and wondered whether a smoker's success or failure on the program could in any way be explained by whether or not the smoker's partner was also a smoker and, if the partner was a smoker, whether they were also attempting to give up smoking.

The organisation contacted a number of people ("subjects") who had been on the program and who had partners. (To avoid double counting, if both subject and partner had been on the program only one of the two would feature as a subject.)

The subjects were asked if they felt that the program had been successful or not, if their partner was also an active smoker at the time and, if so, whether the partner was also trying to give up smoking.

From the responses of 107 people the following table was created.

		Subject's opinion of the program.	
		A success	Not a success
Smoking status of partner.	Not a smoker.	30	20
	Smoker trying to give up.	15	6
	Smoker not trying to give up.	18	18

The organisation operated the stop smoking program in two cities, City A and City B, and wondered if the success rate differed from one city to the next. The organisers knew that of the 107 people who responded to the previous questions, 48 were from people who did the program in city A and, of these, 29 said they considered the program a success.

Is a smoker's opinion of whether the program was a success or not associated with the smoking "status" of their partner?

Was the program in one city achieving greater success than in the other city?

Can the organisers conclude that people attempting to give up smoking have a greater chance of success by being on the program than if they are not on the program?

Investigate. Comment. Write a report.



21. Depending on a person's sleep-wake patterns and preferred activity time throughout the day, they may be classified as a morning, afternoon or evening person. These categories indicate when the person feels most alert and performs their best at mental and physical tests. These time of day variations potentially have implications for society. For example, a morning type person, who would typically perform less well in the evening, may be well advised to avoid night duty. Might the classification in any way be associated with gender? If so, perhaps workplaces that feature significantly more people of one gender than the other should start earlier, or perhaps later, in the day.

Might the classification in any way be associated with age? If so, perhaps the start time at schools should be adjusted accordingly. If teenagers were to be found to be morning people whilst younger children were found to be afternoon people should the respective school start times reflect this?

For the purposes of this question let us suppose that an investigation into such "morningness/afternoon/eveningness" classification led to the following frequency tables, the entry in each cell of the tables indicating the number of people in that category.

	Morning	Afternoon	Evening	Totals
Teenagers	6	44	26	76
20 to 39	23	63	19	105
40 to 59	21	55	14	90
60 and over	23	35	7	65
Totals	73	197	66	336

	Morning	Afternoon	Evening	Totals
Female	42	112	35	189
Male	31	85	31	147
Totals	73	197	66	336

Investigate, comment and write a report based on this data.

Collect, organize and present data, and write a report, to consider one of the following questions (or a different question negotiated between you and your teacher).

Is the male/female balance amongst the top achievers in a school influenced by the year group being considered?

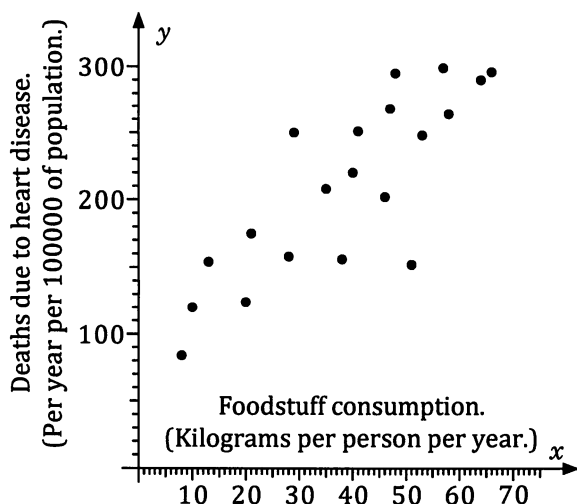
Is the mode of transport chosen to get to school explained to some extent by the age of the school student?

Does the male/female balance of teachers within a subject area change much as we move across subject areas? E.g. English, Mathematics, Phys Ed, etc.

Is there an association between opinion towards capital punishment (in favour of it / no opinion about it / against it) and gender?

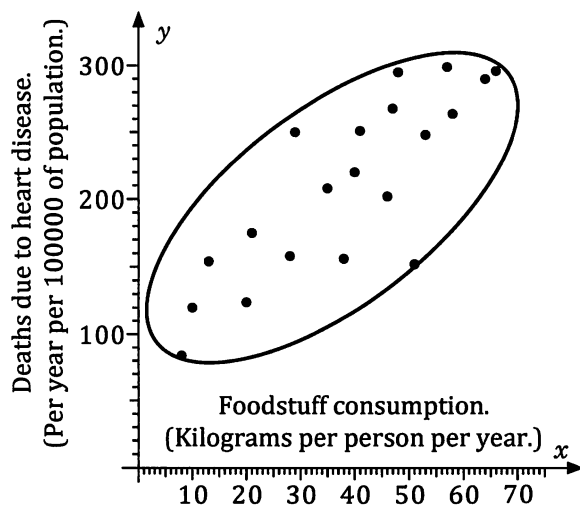
Scattergraphs.

Let us consider again the situation at the beginning of this chapter which involved an investigation into whether countries with higher levels of consumption of a particular foodstuff tended to have higher death rates due to heart disease. We were given foodstuff consumption figures (in kilograms per person per year) and the numbers of deaths due to heart disease (in deaths per year per 100000 of population). Both of these variables, foodstuff consumption and deaths due to heart disease, involve *numerical* data, rather than the categorical data we have been considering more recently. Hence, rather than a two way frequency table, a *scattergraph*, with one variable on one axis, the other on the other axis and the data for each country represented by a point on your graph, becomes the more suitable form of display. Such a scattergraph, shown below, allows the *scatter* of points to be considered.



Notice that the points tend to form an oval shape from lower left to upper right, as shown on the right.

This indicates that in general, countries with a lower consumption of the foodstuff tend to have the lower death rate due to heart disease and those with the higher consumption of the foodstuff tend to have the higher death rate due to heart disease. There seems to be an **association**, or **correlation**, between the foodstuff consumption and the death rate due to heart disease.



Notice the use of phrases like *in general* and *tend to* in the previous paragraph.

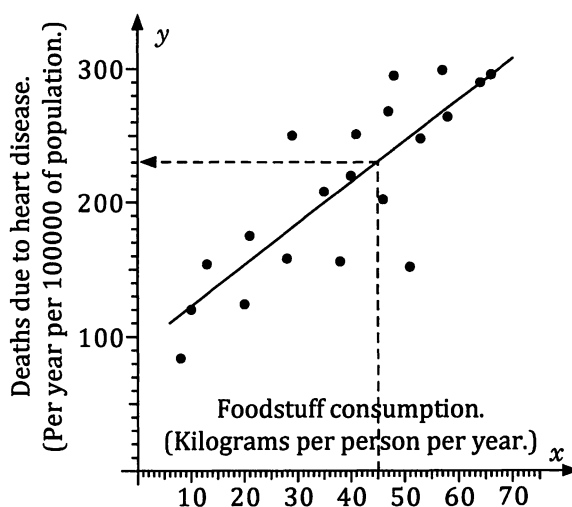
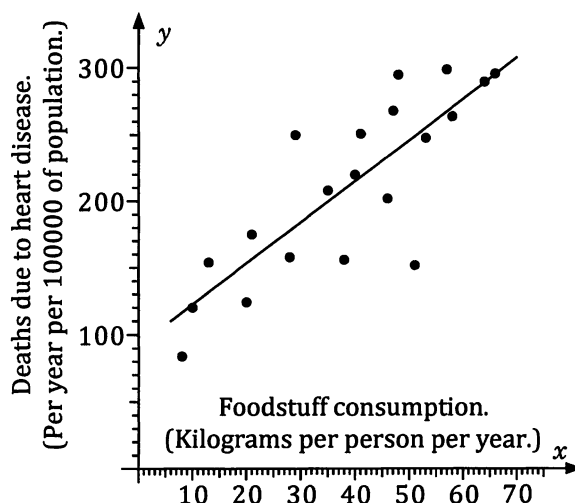
The statement attempts to summarise an underlying pattern or *trend* suggested by the points. Not every individual will necessarily fit the trend, but we can comment on the general behaviour that the points suggest.

We could attempt to draw a straight line that seems to "best fit" this data, as shown on the right.

In this way we fit a **linear model** to the situation.

We could then use this model to make predictions. For example:

Using our **line of best fit** we could suggest that a country with an intake of 45 kilograms per person per year of the foodstuff, might have a heart disease death rate of approximately 230 persons per year per 100000 of population (see the dotted lines in the scattergraph below).



Note: In this foodstuff consumption / death rate due to heart disease situation we are investigating whether the death rate due to heart disease can be explained by the consumption of a particular foodstuff. I.e., is the death rate due to heart disease in any way associated with the amount of the foodstuff consumed? The foodstuff consumption rate is the **explanatory variable** (or **independent variable**), and the heart disease death rate the **response variable** (or **dependent variable**). We tend to plot the explanatory variable, or independent variable, on the horizontal axis and the response variable, or dependent variable, on the vertical axis.

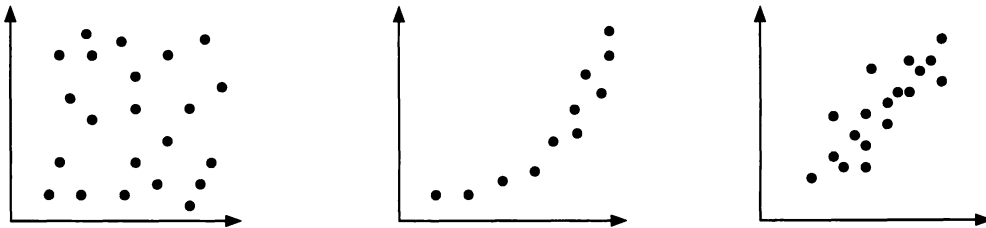
If it is not clear which variable is which, then plot on the horizontal axis the data that is recorded first and plot on the vertical axis the data that is recorded second.

If we are investigating how something changes over *time* one of the variables will be *time* and we have what we call **time series data**. The time variable is the independent variable and is plotted on the horizontal axis. Time series data will be considered in greater detail in *Mathematics Applications Unit Four*.

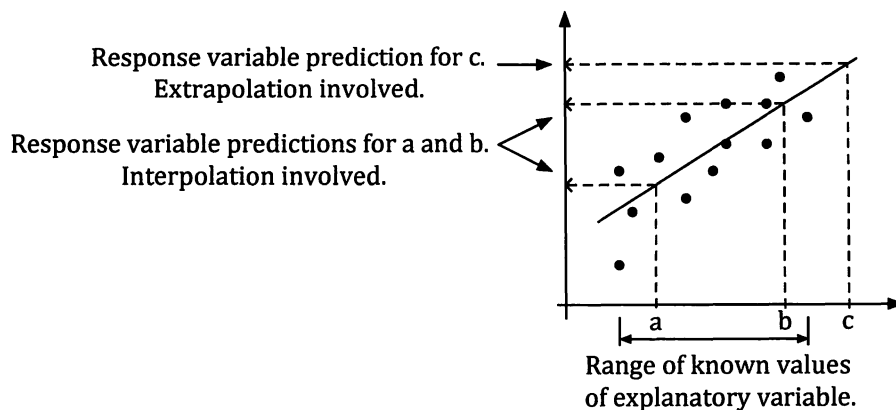
Reliability of our predictions.

When considering just how reliable our prediction might be, the things we need to consider are:

- Does the distribution of points on the scattergraph suggest that using a linear model, or perhaps some other model, is a good idea? For example, three scattergraphs are shown below. In one the distribution of points suggests a linear model would be appropriate, in one the distribution of points could be sensibly summarised using a curve and in one the distribution of points shows no particular pattern. Which graph shows which aspect?

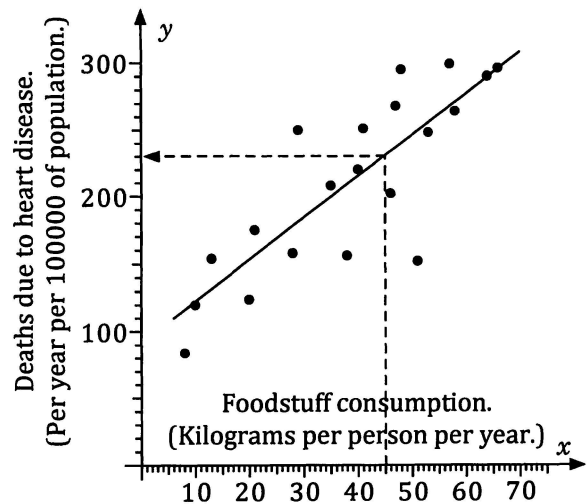


- How many points does our scattergraph involve? The more we have the more confidence we can have in any model they suggest. (After all a scattergraph with just two points fits a linear model perfectly but this could not be expected to give a reliable indicator of trend without the evidence of more points suggesting the suitability of a linear model.)
- How closely do the points fit a straight line? The closer the points lie to a straight line the better the linear model will be and the more confidence we can have in the likely accuracy of our prediction.
- Are we making a prediction for a value that lies within the spread of our known values or is it for a value beyond our known values? I.e., does our prediction involve **interpolation** (*between* the given points) or **extrapolation** (*beyond* the given points). Predictions involving the latter are likely to be less reliable, our confidence in the prediction decreasing the further we are from known points. We often do tend to extrapolate beyond known data points, for example weather forecasts predict the weather in a few days time based on the weather patterns of past days, but the further ahead we are predicting the less certain we are likely to be.



Consider again our previous prediction that for a country with a consumption of a particular foodstuff of 45 kg per person per year, the death rate due to heart disease would be approximately 230 deaths per year per 100 000 people. We could be reasonably confident in this prediction because:

- The graph does suggest a linear model could well be appropriate.
- The graph involves a reasonable number of points.
- The figure of 45 kg per person per year is within the values covered by the given data, i.e. **interpolation** is involved.



However, we could not be so confident if, for example, we continued our "line of best fit" beyond the given data to predict the heart disease death rate for a foodstuff consumption figure of 90 kg per person per year for example. In this case **extrapolation** would be involved because we would be going beyond the range of the known data points.

Exercise 1C

1. CAR VALUES

For the 14 cars of a particular make and model advertised in the *Motors for Sale* section of newspaper one weekend the asking price and age of the cars were as follows:

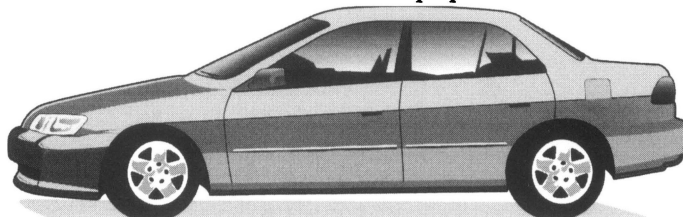
Age (yrs)	13	3	1	7	4	2	11
Price	\$6500	\$19300	\$20988	\$9900	\$16727	\$19990	\$5200
Age (yrs)	6	2	11	1	6	4	9
Price	\$13500	\$16990	\$6250	\$21990	\$12500	\$15500	\$9900

Display these figures as a scattergraph with the age on the horizontal axis and the asking price on the vertical axis.

Comment on any relationships shown between the variables and whether a linear (i.e. straight line) model would be suitable.

Draw on your graph the straight line you feel best summarises the data.

Use your straight line to suggest what a likely price was for a 5 year old car of this make and model at the time of the weekend paper.



\$21990

2. EXAM RESULTS

Fifteen Mathematics students sat a mid-year exam and an end-of-year exam. The results they gained were as follows:

Initials	LA	TB	JD	TE	FG	LG	IH	GL	KL	AL	BR	JS	RS	PT	LV	Out of
Mid-Year	40	34	35	32	36	64	69	63	55	29	59	45	68	66	51	92
End-of-Year	50	69	75	68	64	89	116	98	103	46	82	61	121	88	98	135

- Plot a scattergraph of these results with mid-year results on the horizontal axis and end-of-year results on the vertical axis.
- Comment on any trends shown and whether a linear model would be a reasonable one from which to make estimates.
- Include on your graph what you consider to be the "line of best fit".
- Another student scored 50 out of 92 in the mid-year exam but was hospitalised at the end of the year due to a road accident and hence missed the end-of-year exam. Use your line of best fit to estimate an end-of-year exam mark for this student.

3. SPHERES

A school science experiment involved students in estimating the volumes of certain spheres. The students used graduated beakers containing water and noted the volume of water displaced when each sphere was immersed in the water. The students also measured the diameter of each sphere.

One student recorded the following results:

Sphere	A	B	C	D	E	F	G	H
Diameter	2 cm	3 cm	4 cm	5 cm	6 cm	7 cm	8 cm	9 cm
Volume	5 cm ³	15cm ³	35cm ³	65cm ³	115cm ³	180cm ³	270cm ³	380cm ³

- Represent these paired values as points on a graph with diameter on the horizontal axis and volume on the vertical axis.
- Would a linear model be a good choice for this data?

4. BIRTHDATES

Thirty two year eight students were asked to state their birth dates in the form "day of month/month". Thus a response of 5/11 means the 5th of November. The responses for the 32 students are given below.

13/3	13/6	19/5	24/7	24/4	19/3	15/6	11/4
29/1	15/2	10/3	13/12	3/10	18/2	3/1	15/8
1/5	15/3	16/11	25/12	4/7	25/5	29/11	4/12
20/9	16/6	18/6	20/10	30/9	4/3	24/4	24/1

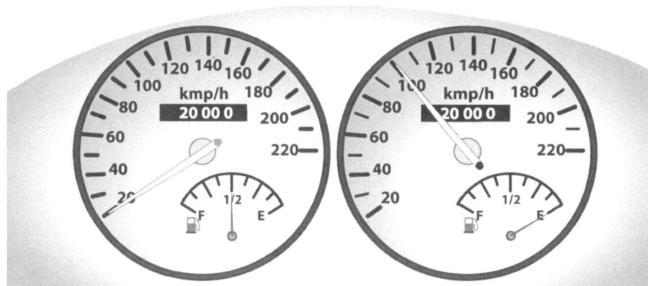
Construct a scattergraph of this data with the day of the month (1, 2, 3, ..., 31) on the horizontal axis and the month (1, 2, 3, ..., 12) on the vertical axis. Comment on any relationships shown.

5. FUEL CONSUMPTION

Twenty cars were tested for fuel consumption when driving in the city and for highway driving. The figures obtained were as shown below.

	Litres per 100 km		Engine Size in Litres
	City Cycle	Highway Cycle	
Vehicle 1	9	6.8	1.8
Vehicle 2	8	6	1.6
Vehicle 3	9.5	6.6	2.0
Vehicle 4	10.5	8	2.5
Vehicle 5	6.4	5.2	1.3
Vehicle 6	12	7.6	4.0
Vehicle 7	14	9	5.0
Vehicle 8	6.8	5	1.3
Vehicle 9	8.5	6.4	1.6
Vehicle 10	8	6.6	1.8
Vehicle 11	10	6.4	2.2
Vehicle 12	8.5	5.6	2.0
Vehicle 13	11	7	3.8
Vehicle 14	14	8.5	5.0
Vehicle 15	14	9	5.0
Vehicle 16	7.2	5.2	1.5
Vehicle 17	10	6.2	2.0
Vehicle 18	10	7.2	3.0
Vehicle 19	10.5	7	2.6
Vehicle 20	10.5	6.6	2.4

- Plot a graph with the highway cycle fuel consumption figures on the horizontal axis and the city cycle fuel consumption figures on the vertical axis. Comment on any relationships indicated by the graph.
- Plot a graph with the engine size on the horizontal axis and the city cycle fuel consumption figures on the vertical axis. Comment on any relationships indicated by the graph.

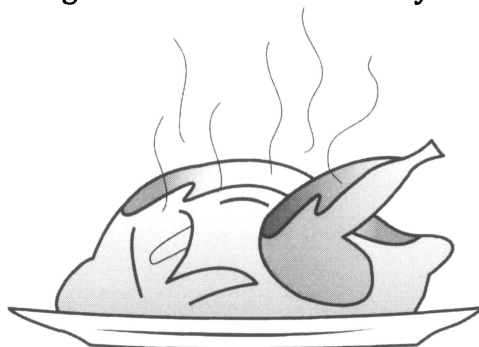


6. COOKING THE MEAT

A cookery book gave the cooking time for a particular joint of meat according to the weight of the joint. The cooking times and weights were as shown in the table below.

Weight	1 kg	1.5 kg	2 kg	2.5 kg	3 kg	3.5 kg	4 kg
Time	1 hour	1 hour 20 mins	1 hour 40 mins	2 hours	2 hours 20 mins	2 hours 40 mins	3 hours

Plot a scattergraph of these figures and comment on any relationships shown.



7. AIRFARES

Let us suppose that the cost of a return flight from Perth to various places in Australia flying business class, together with the distance from Perth to each of these places are as shown in the following table.



Flight	Cost	Distance (One way, by air.)
Perth - Adelaide - Perth	\$2770	2 120 km
Perth - Brisbane - Perth	\$3500	3 611 km
Perth - Darwin - Perth	\$3010	2 651 km
Perth - Melbourne - Perth	\$3070	2 707 km
Perth - Sydney - Perth	\$3140	3 284 km

- (a) Plot a scatter graph of these figures with distance on the horizontal axis and cost on the vertical axis and comment on any relationships shown.

Assuming suitable flights did exist, use your graph to suggest prices for the following return flights, flying business class, and comment on the likely reliability of your estimates.

- (b) From Perth to a place in Australia that is 2 500 km away by air,
 (c) From Perth to a place overseas that is 15 000 km away by air.

8. BREAD ON SPECIAL

The manager of a bread shop monitors the number of loaves of a particular size and style they sell each Saturday morning for eight weeks, and the price they charge for a loaf. The data collected was as shown in the table shown below.

Week	1	2	3	4	5	6	7	8
Price	\$2.70	\$2.35	\$3.05	\$2.60	\$2.50	\$2.95	\$3.20	\$2.80
Loaves sold	342	384	277	331	382	280	254	328

Plot a scattergraph that would allow consideration of how the price per loaf and the number of loaves sold might relate to each other and comment on any relationship shown.

In week 9 the baker plans to charge \$2.90 per loaf. How many would you advise him to make if he wants to sell as many as possible but does not want any left?

9. MEMORY

A list of twenty words was read to a group of 28 students who were then asked to recall as many of the words as they could one minute later. The words were read out again and recall was tested again after 15 minutes. The paired scores achieved in these two memory tests are shown below.

Initials	SA	PB	CC	TC	VC	BD	KD	LF	GG	LH	NH	SH	TH	KL
1st Score	9	15	11	11	13	10	12	10	11	13	9	7	12	11
2nd Score	15	18	14	15	15	17	18	10	11	17	10	9	17	15

Initials	TL	DM	FN	KP	TP	GR	MR	DS	NS	TS	JW	IW	SY	SZ
1st Score	10	10	7	7	18	7	9	6	12	9	14	6	14	5
2nd Score	15	12	10	6	20	8	15	9	11	11	17	7	18	7

Draw a scattergraph of these results and comment on any trends shown.

Draw what you consider to be "the line of best fit" and use this line to predict the likely second score for a student with a first score of 8.

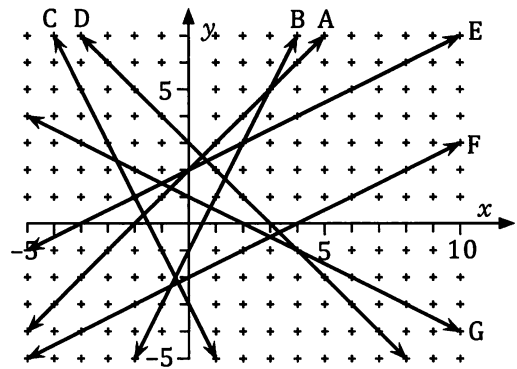
Miscellaneous Exercise One.

This miscellaneous exercise may include questions involving the work of this chapter and the ideas mentioned in the preliminary work section at the beginning of the book.

- Solving equations.
 - If $y = 0.4x + 16.7$, find y when $x = 19.5$.
 - If $y = 4.25x - 17$, find y when $x = 84$.
 - If $y = -7.5x + 307$, find y when $x = 26$.
 - If $y = 5.2x + 3.7$, find x when $y = 128.5$.
 - If $y = -3.25x + 127$, find x when $y = 10$.
- The length of a particular spring, y cm, is found to depend upon the weight, w kg, suspended from it, according to the rule $y = 10w + 25$. Interpret the numbers 10 and 25 in this situation.

3. Copy and complete the following table for the lines A to G shown below.

	Coordinates of y-axis intercept.	Gradient	Equation
A			
B			
C			
D			
E			
F			
G			



4. A survey into the health and fitness of the Australian population sampled over 9000 men and women of certain ages. The data concerning the heights of these people were as follows:

AVERAGE HEIGHTS OF MALE AUSTRALIANS IN HEALTH SAMPLE

Age (yrs)	22	27	32	37	42	47	52	57	62	67
Average Height (cm)	177	177	176	176	176	175	174	173	173	172

AVERAGE HEIGHTS OF FEMALE AUSTRALIANS IN HEALTH SAMPLE

Age (yrs)	22	27	32	37	42	47	52	57	62	67
Average Height (cm)	164	163	163	163	162	163	161	160	160	159

Display these results graphically and comment on any trends shown.

5. Let us suppose that to investigate whether there might be any association between alcohol consumption and the type of crime a person commits, researchers asked 222 people, who had been convicted of particular crimes, to classify themselves by the amount of alcohol they regularly consumed around the time in their lives that they committed the crime. The classifications of alcohol consumption were:

None, Light, Medium, Heavy.

The results led to the following table:

	Usual alcohol consumption.			
	None	Light	Medium	Heavy
Assault	2	6	14	18
Fraud	5	16	10	4
Motor vehicle	11	29	28	17
Theft (not involving assault)	3	22	19	18

Investigate whether the proportions of each type of crime change much as we move across the categories of alcohol consumption. Do the figures suggest there is an association between alcohol consumption and the type of crime committed? If such an association appears to exist write some comments describing its nature.